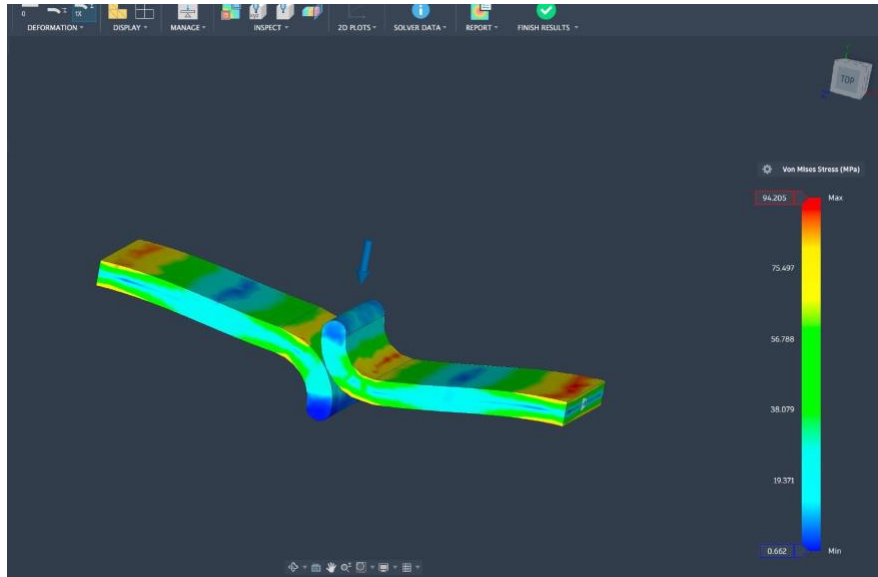


INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

Department of Mechanical Engineering

PROJECT REPORT

Static Stress Analysis of a Bracket under Different Contact Conditions using Autodesk Fusion 360



Von Mises stress distribution on the bracket under a 1000 N load

Submitted by: Ajay Pratap Singh

Roll No: 230103132

Programme: B.Tech, Mechanical Engineering

Software Used: Autodesk Fusion 360 (Simulation Workspace)

1. Objective of the Project

The aim of this project is to understand how the type of contact between two mating parts affects the stress, deformation and overall safety of an assembly, using the Static Stress Analysis tool in Autodesk Fusion 360.

A simple bracket-and-pin type assembly (two components joined at a curved joint) was taken as the test model. The same force and the same fixed support were applied in every case. Only the contact type between the two bodies was changed. This way, the effect of contact type alone could be studied, keeping everything else constant — this is called a controlled comparative study.

In short, the project answers the question: “If two parts of a machine are joined in different ways (bonded, sliding, rough, etc.), how differently will the assembly behave under the same load?”

2. What is Static Stress Analysis?

Static Stress Analysis is a type of simulation used to check how a part or assembly behaves when a constant (non-changing) load is applied to it. It is like a virtual crash-test or a virtual load-test done inside the computer, without actually breaking a physical part.

It helps an engineer find out:

- Strength and weak points of the design
- How the design can be optimised (making it lighter or stronger)
- Whether the design is safe enough for real use (safety verification)

Using this tool before manufacturing saves a lot of time, material and cost, because mistakes are caught on the computer screen instead of in a broken physical prototype.

3. What is a “Contact” in Simulation?

In real assemblies, a machine is rarely made of a single solid part — it is usually two or more separate parts touching each other, such as a pin inside a bracket, a bolt inside a hole, or one bent sheet resting on another. In Fusion 360, the interaction between two such separate bodies is called a Contact.

A contact tells the simulation software how the two touching surfaces should behave under load — whether they can slide over each other, whether they can separate, or whether they are fully stuck together. Defining the correct contact type is very important because:

- It decides how the load actually transfers from one part to the other.
- A wrong contact assumption can give unrealistic results (for example, treating a loose sliding joint as if it were welded).
- Real assemblies almost always involve more than one type of contact.

3.1 Types of Contact (used in this project)

Contact Type	What It Means (In Simple Words)	Real-Life Example
No Contact	The two parts have no relation with each other and behave as if the other part is not there.	Two components placed near each other in an assembly for packaging/space-check only, e.g. a loose tool kept inside a toolbox tray.
Bonded Contact	The two bodies are permanently glued together at the contact surface. No sliding, no separation. Represents welded, glued, or press-fit joints.	A welded steel bracket on a machine frame; a glued mobile phone screen; a press-fitted bearing race inside a housing.
Offset Bonded Contact	Same as bonded contact, but used when the two surfaces have a small gap between them. Fusion 360 still treats them as touching and bonded, ignoring the gap up to a set “Maximum Activation Distance.”	Two laser-cut sheet-metal parts that are meant to be welded but have a small (~1 mm) manufacturing gap before welding; tack-welded joints that are not perfectly flush.
Sliding Contact	The two bodies cannot separate from each other, but their surfaces can slide freely along the contact direction. Friction is not considered.	A drawer channel sliding in its rail; a piston sliding inside a cylinder; a door hinge pin sliding inside its sleeve.
Rough Contact	Behaves like sliding contact but with very high friction, so surfaces do not slide. Unlike bonded contact, the two parts can still separate if pulled apart.	A bolted flange joint held tight by friction under the bolt clamping force; a heavy machine base resting (not fixed) on a rough concrete floor.
Separation Contact	The most realistic contact type. The two bodies can touch, slide freely, and separate from each other depending on the load.	A ball bearing rolling inside its outer race; a cam pushing against a follower; two stacked plates that can lift apart when the load reverses.

Table 1: Different types of contact available in Fusion 360 Simulation

4. Project Set-up (Model, Load and Constraints)

To compare the contact types fairly, the exact same model, material, load and support were used in all four studies. Only the contact definition between the two bodies was changed each time.

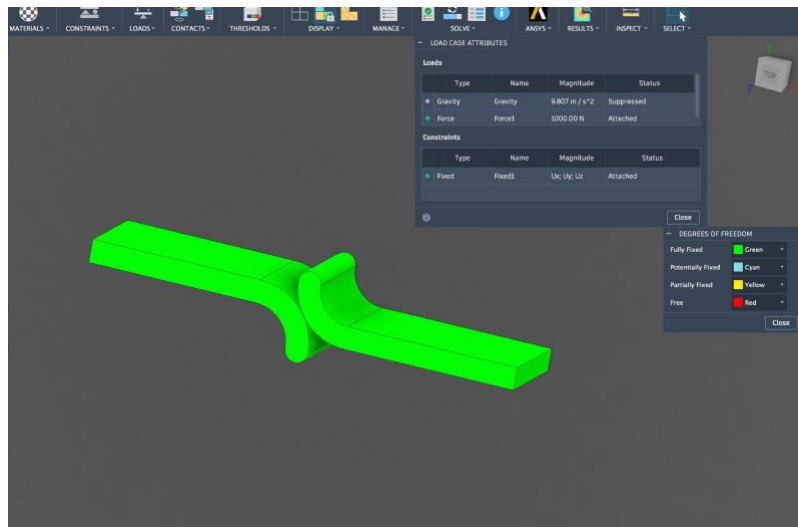


Fig. 1: Load case used for every study — 1000 N downward force and one end fully fixed

4.1 Geometry

The model is a two-piece bracket assembly: a straight lever-like arm that connects into a curved bracket through a common circular/curved joint. The two pieces are separate bodies in the assembly, joined only at this curved region — which is exactly where the different contact conditions were applied and studied.

4.2 Material Selection

The material considered for the study is High-Strength Structural Steel, with the following properties (as used in the Fusion 360 material library):

- Young's Modulus: 200000 MPa
- Poisson's Ratio: 0.29
- Shear Modulus: 77519.38 MPa
- Yield Strength: 344.738 MPa
- Ultimate Tensile Strength: 448.159 MPa
- Density: $7.850 \times 10^{-9} \text{ N} \cdot \text{s}^2/\text{mm}^4$

The material is assumed to be isotropic (same properties in all directions) and linearly elastic (stress is directly proportional to strain, i.e. the part returns to its original shape once the load is removed, as long as the stress stays below the yield strength). Steel was chosen because it is the most common material used for brackets and structural machine parts, and its well-documented properties make the Safety Factor results easy to judge and cross-check.

4.3 Loading and Boundary Conditions

The same loading and boundary conditions were used for every one of the four studies, so that only the contact type varied between them:

- Applied Load: 1000 N (downward force at the free end of the arm)
- Gravitational acceleration: 9.807 m/s^2 (included in the model but kept Suppressed/switched off, so only the 1000 N force acts)
- Constraint: One end of the assembly Fixed ($U_x, U_y, U_z = 0$) — similar to a bracket bolted rigidly to a wall or machine frame
- Analysis type: Static Structural Analysis (constant, non-changing load)
- Safety factor criterion: Yield Strength
- Target factor of safety: 2.0 (a part is considered “safe” if its minimum Safety Factor stays at or above this value)

1000 N (about 100 kgf) was chosen as the applied load because it is a round, realistic, moderate value — large enough to produce a clear, measurable amount of stress and deflection in the part (so the difference between contact types is clearly visible), yet well within the range a small-to-medium steel bracket would normally be expected to carry in service. The exact magnitude matters less than the fact that it was kept identical across all four studies: because the geometry, material, load and support are all unchanged, any difference in the results can be attributed only to the change in contact type — this is what makes the comparison valid.

4.4 Offset — Why “Offset Bonded” was studied, and the value used

The Offset Bonded contact is a special case of a bonded (glued) contact. It is used when two surfaces are supposed to be touching, but in the actual CAD model there is a small manufacturing/modelling gap between them (an “offset”). Instead of treating this small gap as a real air gap (which would make the simulation fail to transfer load, since the parts appear disconnected), Fusion 360 lets the user tell the solver: “treat any gap up to this distance as if the surfaces are directly touching and bonded.”

Maximum Activation Distance used: 1 mm. This means any gap between the two surfaces up to 1 mm was still treated by the solver as a fully bonded joint. If the real gap in the model is larger than this set value, the contact would not activate and the two parts would behave as if disconnected.

This study was included to check how much difference a small real-world assembly gap/misalignment makes to the results, compared to a perfectly touching bonded joint. Interestingly, the results in Section 6 show that the Offset Bonded (1 mm) case gave a noticeably lower Safety Factor (2.477) and higher stress (139.18 MPa) than the plain Bonded case (3.659, 94.21 MPa) — even though both are technically “bonded.” This happens because, over the 1 mm offset, load has to transfer across a small artificial gap rather than a truly continuous surface, which locally raises the stress near the joint. This is a useful, realistic finding: even a “bonded” joint is not as strong as a perfectly mating one if there is a real manufacturing offset involved.

5. Methodology

The following steps were followed for each of the four studies:

- Step 1: Import/model the two-body bracket assembly.
- Step 2: Assign High-Strength Structural Steel material to both bodies.
- Step 3: Apply a Fixed constraint at one end of the assembly.
- Step 4: Apply a 1000 N Force at the free end of the arm.
- Step 5: Suppress the gravity load so only the applied force acts.
- Step 6: Define the contact between the two bodies as Bonded / Offset Bonded / Sliding / Rough (one type per study).
- Step 7: Generate the mesh (the model is divided into small tetrahedral elements for calculation).
- Step 8: Run/Solve the study on the Fusion 360 cloud solver.
- Step 9: Study the results — Safety Factor, Von Mises Stress, Displacement and Contact Force.

This was repeated four times, changing only Step 6 each time, giving four independent studies: Bonded Contact, Offset Bonded Contact, Sliding Contact and Rough Contact.

6. Results

The table below summarises the key results obtained from all four studies. The Safety Factor is calculated using the Yield Strength of the material — a Safety Factor above 2.0 was the target set in the study, meaning the part is not expected to bend permanently or fail.

Contact Type	Min. Safety Factor	Max. Von Mises Stress (MPa)	Max. Displacement (mm)	Max. Contact Force (N)
Bonded Contact	3.659	94.21	0.078	248.71
Offset Bonded (1 mm)	2.477	139.18	0.212	613.15
Sliding Contact	3.089	111.60	0.099	1023.74
Rough Contact	1.395	247.08	0.456	283.97

Table 2: Comparison of results for all four contact conditions

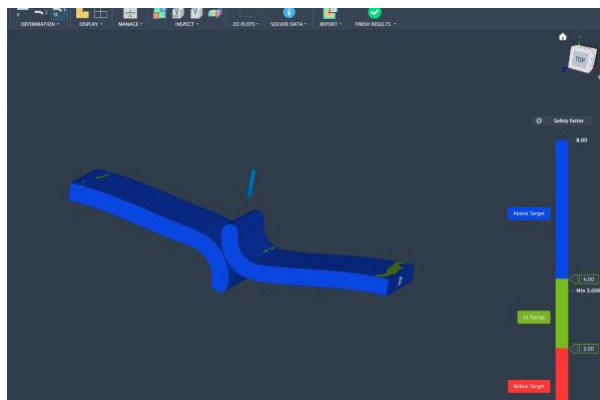
6.1 Result Parameters Used in This Study

Before looking at the individual results, it helps to know exactly what each parameter means in the simulation:

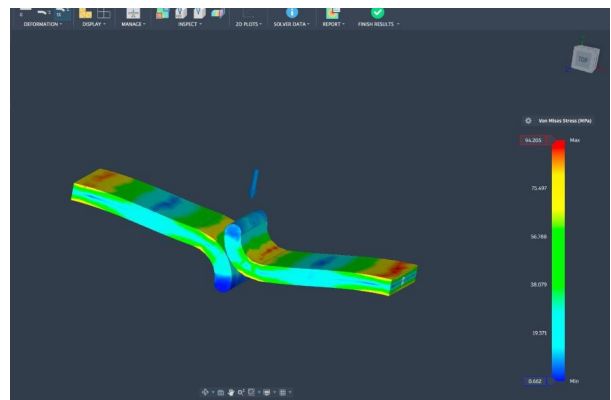
Parameter	What It Means in This Simulation
Safety Factor	How many times stronger the part is compared to what the current load demands, based on the material's Yield Strength. A Safety Factor of 2 means the part can, in theory, take double the applied load before it starts to bend permanently.
Von Mises Stress	A single combined stress value (in MPa) calculated from all the direct and shear stresses at a point, used to judge yielding/failure of ductile materials like steel. It is compared against the Yield Strength to compute the Safety Factor.
Displacement	How far a point on the part physically moves (in mm) from its original position after the load is applied, in any direction. It shows how much the part bends or deflects under the 1000 N force — larger displacement means a more flexible part.
Contact Force	The total force (in N) being transmitted through the contact area between the two bodies. It shows how much of the applied 1000 N load is actually passing from one body into the other at the joint, and helps judge how efficiently the joint shares the load.
Contact Pressure	The contact force spread over the actual contact area, i.e. force per unit area (in MPa) at the touching surfaces. A high contact pressure at a small spot (as seen in the Rough Contact case) indicates a local hot-spot where the material is most likely to yield first.

Table 3: Definition of the result parameters reported in this study

6.2 Bonded Contact



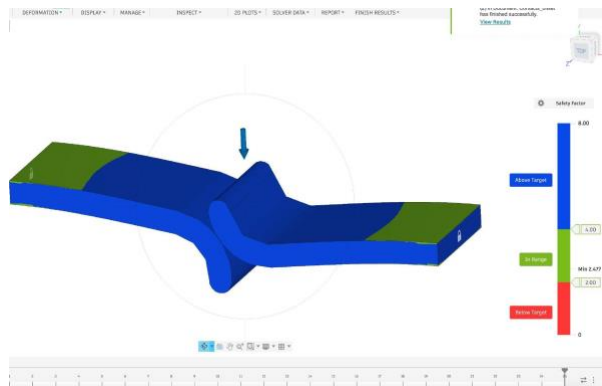
Safety Factor = 3.659 (Strong enough)



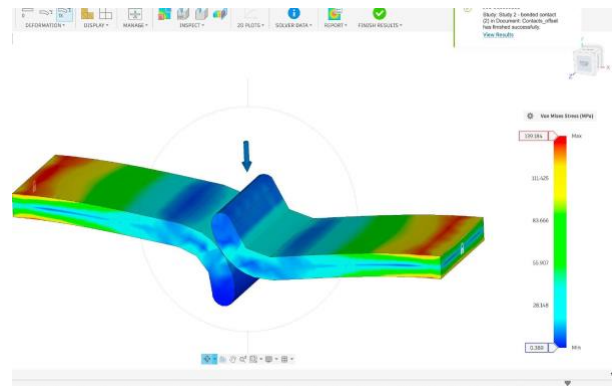
Von Mises Stress, Max = 94.21 MPa

With a bonded contact, the two bodies act almost like a single solid piece. The load transfers smoothly and continuously between them, giving low stress and a comfortably high safety factor.

6.3 Offset Bonded Contact (Maximum Activation Distance = 1 mm)



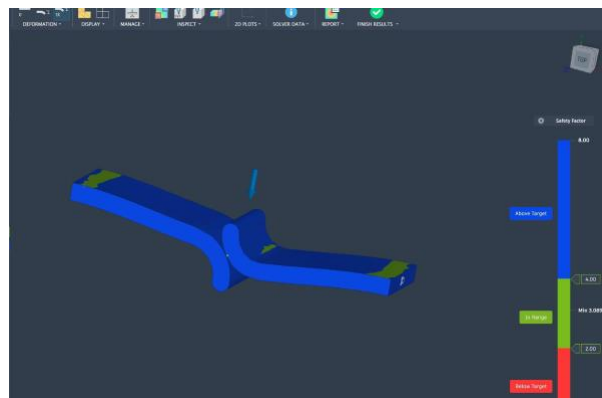
Safety Factor = 2.477 (Strong enough)



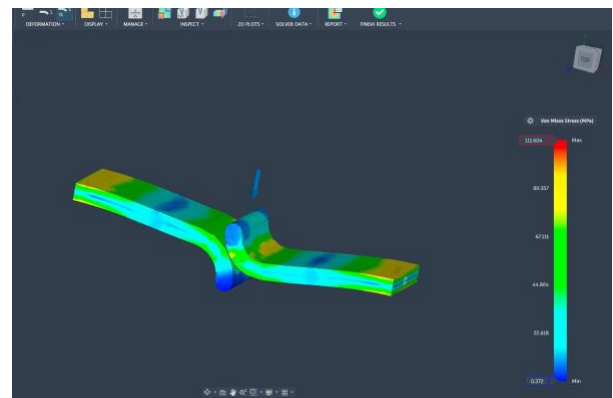
Von Mises Stress, Max = 139.18 MPa

With a 1 mm offset, the joint is still treated as bonded, but stress rises noticeably compared to the plain bonded case, and the Safety Factor drops from 3.659 to 2.477 — still above the 2.0 target, but with a much thinner safety margin. This shows that even a small (1 mm) real-world offset between two “bonded” surfaces can meaningfully reduce a design's strength margin.

6.4 Sliding Contact



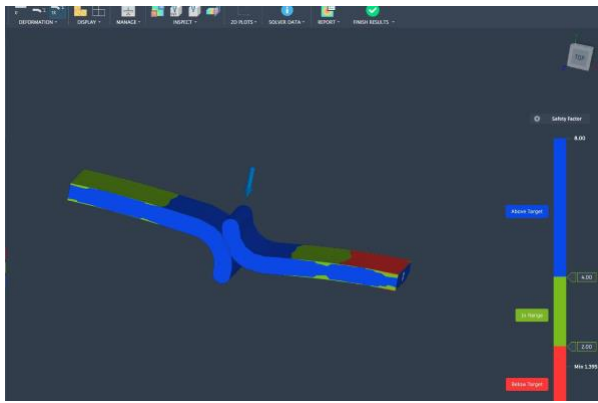
Safety Factor = 3.089 (Strong enough)



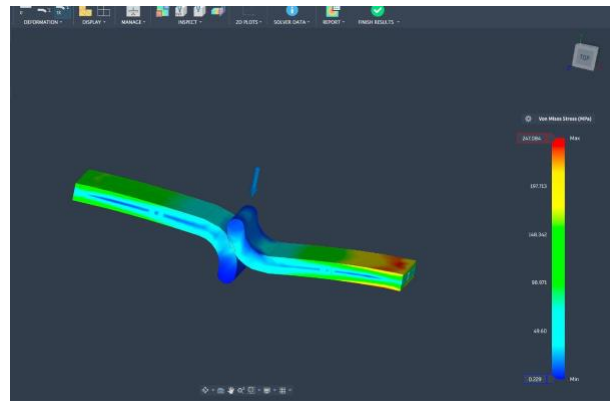
Von Mises Stress, Max = 111.60 MPa

Since the surfaces are now free to slide over each other, the joint is less rigid than a bonded joint. This causes higher local stress and a noticeably larger contact force (about 1024 N) at the joint, although the part is still within the safe range.

6.5 Rough Contact



Safety Factor = 1.395 (Not strong enough)



Von Mises Stress, Max = 247.08 MPa

This is the most critical case. Because the two bodies are allowed to separate from each other (unlike bonded or sliding contact), the load path becomes concentrated over a much smaller contact area. This causes the stress to rise sharply (247 MPa, more than double every other case) and the Safety Factor drops below the target value of 2.0 — meaning the design is predicted to fail under this contact assumption.

7. Observations and Inference

Ranking the four studies from safest to most critical (highest to lowest Safety Factor):

- 1. Bonded Contact — Safety Factor 3.659 (safest)
- 2. Sliding Contact — Safety Factor 3.089
- 3. Offset Bonded Contact (1 mm) — Safety Factor 2.477
- 4. Rough Contact — Safety Factor 1.395 (fails the 2.0 target)
- The contact type has a very large effect on the results, even though the geometry, material and load were exactly the same in every study.
- Plain Bonded contact gave the safest result, because the two bodies act as one continuous solid — there is no gap or relative motion anywhere, so load transfers with the least resistance.
- Sliding contact came second-safest, even though the surfaces can move relative to each other, because the two bodies still stay in full, continuous surface contact (they just cannot separate) — this keeps the load spread over a wide area.
- Offset Bonded (1 mm) came third, below sliding contact. This was an interesting and non-obvious result: adding even a small 1 mm offset to a bonded joint made it weaker than a perfectly-mating sliding joint, because load has to bridge across the small gap, concentrating stress near the joint.
- Rough contact gave the worst (lowest) safety factor and the highest stress of all four cases, because it is the only contact type here that allows the two bodies to separate under load — this concentrates the entire load onto a much smaller contact patch.
- Overall takeaway: choosing the correct, realistic contact type — and correctly modelling any real gaps or offsets — is just as important to a safe design as choosing the correct material or load. Assuming a loosely-fitting real joint is “fully bonded” can give a dangerously over-optimistic result.

8. Real-World Applications

This kind of contact-based static stress study is used in many areas of mechanical and product design, for example:

- Bolted and pinned joints — brackets, hinges, clevis-pin joints in machines and automobiles.
- Bearing and shaft assemblies — where sliding/rough contact realistically represents rotating or sliding parts.
- Sheet-metal and welded assemblies — bonded contact represents welded or riveted joints in vehicle chassis, machine frames, and support structures.
- Gears, cams and mating machine parts — where surfaces slide and separate under changing loads.
- Product design and prototyping — checking a design digitally before spending money on a physical prototype, reducing development time and cost.
- Aerospace and automotive structures — where multiple riveted/bolted panels must be checked for how loads pass between them.

9. Conclusion

This project used Autodesk Fusion 360's Static Stress Analysis tool to study how four different contact conditions — Bonded, Offset Bonded, Sliding and Rough — affect the stress, deformation, and safety factor of a bracket assembly under an identical 1000 N load.

The results clearly show that a contact definition is not just a simulation setting but an engineering decision that must match the real physical joint. Bonded and Offset Bonded contacts represent rigid, permanent joints and gave the safest results. Sliding contact represents a freely-sliding joint and gave intermediate results. Rough contact, which allowed the parts to separate, gave the most critical (least safe) results and even failed the target safety factor of 2.0.

Overall, the project reinforced the importance of correctly modelling how parts interact in an assembly, and demonstrated a complete, practical simulation workflow — from setting up geometry, material, loads and constraints, to defining contacts, meshing, solving, and interpreting Safety Factor, Stress, Displacement and Contact Force results.

10. Key Skills Gained

- Hands-on experience with Autodesk Fusion 360 Simulation workspace (Static Stress study).
- Understanding of Finite Element Analysis (FEA) basics — meshing, boundary conditions, and solving.
- Practical knowledge of different contact types and when to use each one in a real design.
- Ability to interpret engineering results such as Von Mises Stress, Safety Factor, Displacement and Contact Force.
- Comparative / controlled-variable analysis approach to engineering problem solving.